Lab 1 calculations

Subnet 192.168.5.0/24 network to provide sufficient addressing for each LAN, + Point to Point connection between R1 + R2.

LAN 1 : 45 host

LAN 2 : 64 host

LAN 3 : 14 host

LAN 4 : 9 host

R1 ↔ R2 : 2 host

1. VLSM start with the subnet with the most host

LAN2 : 64 host

192.168.5.0/25 ← Network Address

- Find the Pre-fix length

2	1
4	2
8	3
16	4
32	5
64	6
128	7

we need 7 host bits

32 - 7 = /25 prefix length

- last dotted decimal in binary

net | host

0 | 0000000

$$\begin{array}{r} 255 \\ -128 \\ \hline .127 \end{array}$$

Broadcast Address

192.168.5.127/25

2. continue with the next largest LAN.

LAN1 = 45 hosts

* use the next address after LAN2 Broadcast address. this will be LAN1 network Addr.

192.168.5.128/26 ← network Addr.

- Find most sufficient prefix length

2	1
4	2
8	3
16	4
32	5
64	6

6 host bits = 64 Addr. - 2 = 62 usable addresses

32 - 6 = /26 prefix length

net | host

.10|000000

192.168.5.191 ← broadcast Addr.

3. continue with the next largest LAN.

LAN3 = 14 hosts

* use the next address after LAN1 Broadcast address. this will be LAN3 network Addr.

192.168.5.192 /28 ← network Address

• find most sufficient prefix length

2 1
4 2
8 3
16 4

4 host bits = 16 addresses = 14 usable

32 - 4 = /28 prefix length

network | host

.11000000

128
64
192
32
224

.207

192.168.5.207/28 ← Broadcast Addr.

next LAN LAN4 = 9 host

will need 4 hosts bits = 16 - 2 = 14 usable

32 - 4 = /28

192.168.5.208/28 ← network Addr